STATS 200A Discussion 5

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27 October 2026

$\mathbb{P}_0 \sim N(0,1), \lambda << \mathbb{P}_0$
$= \int_A f_{N(0,1)}\, dx = \mathbb{E}_{N(0,1)}(A)$
$\mu << \nu, A : \nu(A) = 0 \Rightarrow \mu(A) = 0$

$$\begin{aligned}
\mathcal{R} &= \bigcup_{k=1}^{\infty} [-k.k] \\
A &= A \cap \mathbb{R} = A \cap \left(\bigcup_{k=1}^{\infty} [-k, k] \right) \\
&= \bigcup_{k=1}^{\infty} (A \cap [-k, k]) \\
&\quad\ \ A_k \\
\mathbb{P}_0(A_k) &= \int_{A_k} \frac{e^{-x^2/2}}{\sqrt{2\pi}}\, dx \\
&\leq \int_{A_k} \frac{e^{-k^2/2}}{\sqrt{2\pi}}\, dx \\
&= \frac{e^{-k^2/2}}{\sqrt{2\pi}} \int_{A_k} \mathbf{1}\, dx \\
&= \frac{e^{-k^2/2}}{\sqrt{2\pi}} \lambda(A_k) = 0 \\
\mathbb{P}_0(A_k) &= 0 \\
A &= \bigcup_{k=1}^{\infty} A_k \Rightarrow \mathbb{P}_0(A) \leq \sum_{k=1}^{\infty} \mathbb{P}_0(A_k) = 0
\end{aligned}$$

$\mathbb{P}_\theta \sim N(\theta, 1)$

$$\nu = \frac{\partial \mathbb{P}_\theta}{\partial \mathbb{P}_0} = \frac{\frac{1}{\sqrt{2\pi}} \exp\left(\frac{-(x-\theta)^2}{2}\right)}{\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)} = \exp\left(x\theta - \frac{\theta^2}{2}\right)$$

$$\begin{aligned}
\mathop{\mathbb{E}}_{x \sim N(0,1)} (\nu(x)) &= \int_{\mathbb{R}} \exp\left(x\theta - \frac{\theta^2}{2}\right) \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}\, dx \\
&= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{x\theta - \frac{\theta^2}{2} - \frac{x^2}{2}}\, dx \\
&= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \exp\left(-\frac{x^2 + \theta^2 - 2x\theta}{2}\right) dx \\
&= \theta
\end{aligned}$$

$$f_{X,Y}(x, y) = ce^{-2x} \mathbf{1}_{0<y<x<\infty}$$

$X, Y \overset{iid}{\sim} F$

$$\mathbb{E}(X \mid G) = Z\text{random variable} \begin{cases} Z : G - \text{measurable} \\ \mathbb{E}\,|Z| < \infty \\ A : G - \text{measurable}\ \mathbb{E}(X \cdot \mathbf{1}_A) = \mathbb{E}(Z \cdot \mathbf{1}_A) \end{cases}$$

Z is G measurable if

$$B \subseteq \mathbb{R} \quad Z^{-1}(B) = \{\omega \in \Omega \mid Z(\omega) \in B\} \in G$$

$X : \Omega \to \mathbb{R}, \mathcal{F}_0\{\omega, \emptyset\}$
$\mathbb{E}(X \mid \mathcal{F}_0) = \mathbb{E}(X) = 2 = Z$
$Z^{-1}(\{2\}) = \Omega, Z^{-1}(\mathbb{R}\backslash\{2\}) = \emptyset$
Hence $Z = \mathbb{E}(X)$ is $\mathcal{F}_0$ measurable.
$\mathbb{E}(X \cdot \mathbf{1}_A) = \begin{cases} \mathbb{E}(X \cdot \mathbf{1}_\Omega) = \mathbb{E}(X) = Z = \mathbb{E}(Z \cdot \mathbf{1}_\Omega) \\ \mathbb{E}(X \cdot \mathbf{1}_\emptyset) = 0 = \mathbb{E}(Z \cdot \mathbf{1}_\emptyset) \end{cases}$

$$\mathbb{E}(X \mid X + Y) = \mathbb{E}(X \mid \sigma(X + Y))$$
$$= \begin{cases} \mathbb{E}(X) + \mathbb{E}(Y) & \checkmark \\ \mathbb{E}(X) & \checkmark \\ X + \mathbb{E}(Y) & \times \\ Y + \mathbb{E}(X) & \times \\ \frac{X+Y}{2} & \checkmark \\ 2X + 3Y & \times \end{cases}$$

Theorem 5.1 (Doob-Dynkin)

$$\mathbb{E}(X \mid \sigma(Y)) = g(Y)$$

$\mathbb{E}(X \mid X + Y) = \frac{X+Y}{2}$
$\frac{X+Y}{2} \in \sigma(X + Y), \mathbb{E}\left|\frac{X+Y}{2}\right| \leq \mathbb{E}\,|X| + \mathbb{E}\,|Y| < \infty$
$A \subseteq \sigma(X + Y), \mathbb{E}(X\mathbf{1}_A) = \mathbb{E}\left(\frac{X+Y}{2}\mathbf{1}_A\right)?$
$\mathbb{E}(X\mathbf{1}_A) = \mathbb{E}(Y\mathbf{1}_A)$
$\mathbb{E}(\frac{X}{2}\mathbf{1}_A) = \mathbb{E}(\frac{Y}{2}\mathbf{1}_A)$
$\mathbb{E}(\frac{X+Y}{2}\mathbf{1}_A) \overset{?}{\Leftarrow} 2\mathbb{E}(\frac{X}{2}\mathbf{1}_A) = \mathbb{E}(\frac{Y}{2}\mathbf{1}_A) + \mathbb{E}(\frac{Y}{2}\mathbf{1}_A)$

$$\int_A x\, d\mathbb{P} = \int_A \mathbb{P}_X(x)\, dx = \int_A \mathbb{P}_Y(x)\, dx = \int_A Y\, d\mathbb{P} = \mathbb{E}(Y\mathbf{1}_A)$$